

Respostas Atividade Aula 006

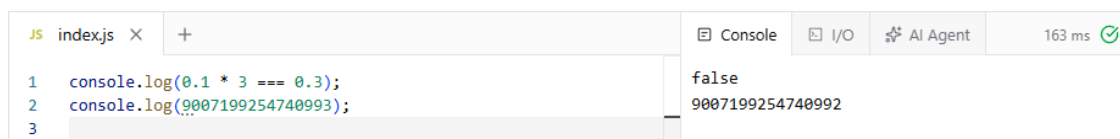
1 - JavaScript

Código:

```
console.log(0.1 * 3 === 0.3);  
console.log(9007199254740993);
```

Saída:

```
false  
9007199254740992
```



Saída js

Erro: O erro é percebido na execução. Dado que o valor esperado é diferente do real. E acontece um arredondamento.

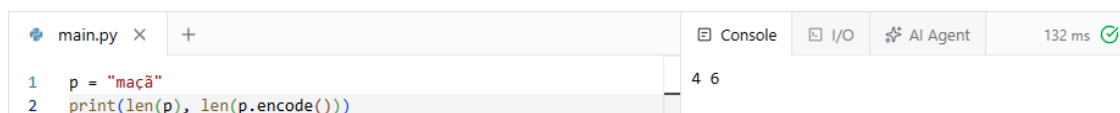
2 - Python

Código:

```
p = "maçã"  
print(len(p), len(p.encode()))
```

Saída:

4 6



Saída py

Erro: Nenhum erro é detectado. O que acontece é que caracteres especiais possuem um tamanho maior.

3 - Go

Código:

```
var b byte = 255  
b++  
fmt.Println(b)
```

Saída:

0

```
main.go X
1 package main
2 import "fmt"
3
4 func main() {
5
6     var b byte = 255
7     b++
8     fmt.Println(b)
9
10 }
```

Console I/O AI Agent 245 ms ✓

0

Saída go

Erro: O byte acaba retornando (Volta), o erro nunca é detectado.

4 - Java

Código

```
int[] v = new int[3];
System.out.println(v[0]);
System.out.println(v[3]);
```

Saída:

Exception in thread "main" 0 java.lang
ArrayIndexOutOfBoundsException: Index 3 out of bounds for length 3 at Main.main(Main.java:8)

```
Main.java X +
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         int[] v = new int[3];
7         System.out.println(v[0]);
8         System.out.println(v[3]);
9
10     }
11 }
```

Console I/O AI Agent Exit 1

Exception in thread "main" 0
java.lang.ArrayIndexOutOfBoundsException:
Index 3 out of bounds for length 3
at Main.main(Main.java:8)

Debug this error

Saída java

Erro: O erro é detectado na compilação. O Java utiliza um index a partir de zero, logo o maior num array de 3 itens, é o index 2. O index 3 está fora de alcance (Out of bouds).

5 - Rust

Código:

```
let s = String::from("oi");
let t = s;
println!("{}", s, t);
```

Saída:

```
error[E0382]: borrow of moved value: `s`
--> HelloWorld.rs:5:23
```

```
3 |     let s = String::from("oi");
  |           - move occurs because `s` has type `String`, which does not i
  |           mplement the `Copy` trait
4 |     let t = s;
  |           - value moved here
5 |     println!("{}", s, t);
  |                  ^ value borrowed here after move
```

help: consider cloning the value if the performance cost is acceptable

```
4 |     let t = s.clone();
  |           +++++++
```

error: aborting due to 1 previous error

For more information about this error, try `rustc --explain E0382`.

```
fn main() {
    let s = String::from("oi");
    let t = s;
    println!("{}", s, t);
}
```

```
error[E0382]: borrow of moved value: `s`
--> HelloWorld.rs:5:23

3 |     let s = String::from("oi");
  |           - move occurs because `s` has
  |           type `String`, which does not implement the
  |           `Copy` trait
4 |     let t = s;
  |           - value moved here
5 |     println!("{}", s, t);
  |                  ^ value borrowed
  |                  here after move
  |
help: consider cloning the value if the
     performance cost is acceptable
4 |     let t = s.clone();
  |           +++++++

error: aborting due to 1 previous error

For more information about this error, try
`rustc --explain E0382`.
```

Saída rs

Erro: Detectado na compilação.

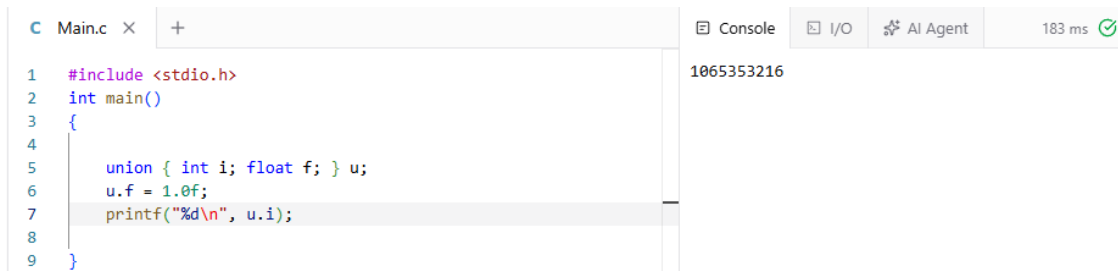
6 - C

Código:

```
union { int i; float f; } u;  
u.f = 1.0f;  
printf("%d\n", u.i);
```

Saída:

1065353216



The screenshot shows a code editor with a file named 'Main.c'. The code defines a union with an integer member 'i' and a float member 'f'. It initializes the float member 'f' with the value 1.0f and then prints the integer member 'i'. The output in the console is 1065353216, which is the bit-level representation of the float value 1.0. The editor interface includes a tab for 'Main.c', a 'Console' panel on the right showing the output, and a status bar indicating a runtime of 183 ms.

```
C Main.c x +  
1 #include <stdio.h>  
2 int main()  
3 {  
4  
5     union { int i; float f; } u;  
6     u.f = 1.0f;  
7     printf("%d\n", u.i);  
8  
9 }
```

Console I/O AI Agent 183 ms ✓

1065353216

Saída c

Erro: O erro nunca é detetado.